Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/11/2005
PI Name:	Glazier, James Alexander Ph.D.		
Project Title:	Comparative Study Of Morphogenesis In 1g And Simulated Microgravity/Subproject B		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture		
PI Email:	glazier@indiana.edu	**Fax:**	FY 812-855-5533
PI Organization Type:	UNIVERSITY	**Phone:**	812-855-3735
Organization Name:	Indiana University		
PI Address 1:	727 E 3rd Street/ Swain West 159		
PI Address 2:	Department of Physics		
PI Web Page:	http://www.biocomplexity.indiana.edu		
City:	Bloomington	**State:**	IN
Zip Code:	47405-7105	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	03/01/2003	**End Date:**	02/28/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	2
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG2-1619		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The goal of this project is to compare fundamental embryological morphogenetic phenomena under normal and zero gravity conditions. Specific areas to be considered include gastrulation, limb development and cell sorting. Tasks include development of biomechanical models for devlopment and experiments under normal and simulated microgravity conditions. These experiments and simulations will be coordinated with supporting and parallel research by Dr. Rusty Lansford (CalTech) and Dr. Gabor Forgacs (U. Missouri, Columbia). Task Objective: Develop protocols necessary for assays of effects of simulated microgravity on cell motility and signaling during developmental processes. Task Significance: Since many simulated microgravity effects on development are quite subtle, it is essential to have a complete range of quantitative techniques to assess changes in cell behavior related to cytoskeletal disruption under		

	microgravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We have discovered a fundamentally new patterning mechanism operating in chick limb beyond classical positional coding and reaction-diffusion. This mechanism has fundamentaly different character and is probably important in a large variety of other developmental processes in other animals. Understanding the mechanisms of development is a key step in developing therapies to treat developmental abnormalities, design regenerative medical therapies and treat degenerative diseases in adults. The microfluidics cell, when further developed will assist in automating and streamlining drug discover and evaluation and may have applications in detecting biotoxins.
Task Progress:	During the 2004 calendar year we developed an extensive set of techniques for analyzing cell behaviors, particularly cell motility, adhesivity, cell shape and differential response to external signals, which we will apply to cells in simulated microgravity conditions in 2005. These cell behaviors are all crucial to normal tissue function, renewal, degeneration and wound healing and are also sensitive indicators of changes in the structure and function of the cytoskeleton. Motility and adhesivity changes show changes to actin function, while changes in cell shape and mechanical rigidity primarily reveal changes in microtubule behavior. A complete repertory of well-established and sensitive techniques is essential if we are to look for the rather subtle cytoskeleton-induced changes in cell behavior we expect to be associated with micro-gravity conditions. Because renovations of the Glazier Laboratory took until summer, 2004, some of this work took place in other researchers' laboratories. The Glazier laboratory purchased and installed a separately funded spinning-disk confocal microscope (McBain/Nikon) for use in cell-tracking and fate-mapping experiments for this project and a separately funded flow cytometer/cell sorter (FACSCaliber) for use in gene expression studies. The chief techniques which we have established are: 1) Micromass culture of limb mesenchymal cells from early stage chick embryos (With Prof. Stuart Newman, New York Medical College). In micromass culture, we dissect and dissociate chick mesenchymal cells and plate them out to observe their condensation into precartilage nodules. This technique assays cell motility, cell-cell signaling and cell differentiation into cartilage. All areas which would reveal subtle changes in cell behavior due to cytoskeletal alterations. Changes in behavior revealed by this technique could help point to potential causes and countermeasures for microgravity-induced cartilage and bone defects. This technique is now running reliably in the Glazier laboratory. 2) Whole-embryo chick culture. In this technique we culture the entire embryo to observe limb development in quasi-in vivo conditions. When combined with cell-tracking and fate-mapping, this technique allows us to investigate the functional determinants of limb formation. This technique is now running reliably in the Glazier laboratory. 3) EC culture of early stage gastrulating whole chick embryos (with Prof. Cornelis Weijer, University of Dundee, Scotland). This technique allows us to look for changes in bulk morphology and cell motility during primitive streak formation. The cell movements during gastrulation are the most dramatic of any during development and even subtle changes in cell motility or adhesion cause dramatic patterning defects. Thus the technique provides a sensitive assay for cytoskeletal modifications due to microgravity. This technique is now running reliably in the Glazier laboratory. Cheng Cui, a graduate student from the Glazier laboratory has joined the laboratory of Dr. Charles Little (University of Kansas Medical Center), where he will continue this work, focusing on the properties and behavior of the extracellular medium. 4) Electroporation (with Prof. Cornelis Weijer, University of Dundee, Scotland). Surprisingly, there was no effective protocol for electroporation of chick embryos to introduce DNA for labeling and over expression experiments. A graduate student (Wei Zeng) worked jointly with Prof. Weijer and Prof. Glazier to develop a reliable electroporation protocol. We intend to use this technique for cell-tracking, cell fate-mapping and to study the effect of over expression of key morphogenetic molecules (e.g. fibronectin and TGF-?) on morphogenesis in cells exposed to microgravity conditions. 5) Transfection using retroviruses (with Dr. Rusty Lansford, California Institute of Technology, co-PI on this NASA grant). A graduate student (Wendy Zhang) spent several months in Dr. Lansford's laboratory learning his retroviral labeling and cell-tracking techniques. These techniques have similar benefits to those of electroporation, but can be used in a greater variety of situations and label a larger fraction of cells within the embryo. She has now returned to the Glazier laboratory to implement these techniques. 6) Microfluidics assays of Chemotaxis and Haptotaxis (with Prof. Stephen Jacobson, Dept. of Chemistry, Indiana University, Bloomington). A graduate student, Dragos Amarie, has developed a series of microfluidics devices which can quantitatively assess cell motility in response to external chemical gradients either in the fluid bathing the cell or in the substrate. This technique provides a sensitive assay for changes in cell motility and in cell-response to external chemical signals. We are currently improving and refining these techniques and conducting preliminary chemotaxis assays on mouse sperm and fish keratocytes (which are standard cell-types for calibrating these devices). 7) HUVEC assays of angiogenesis (with Prof. Michael S. Goligorksy, New York Medical College). These in-vitro assays study the mechanisms of vascular patterning, which is important both during development and during wound healing (See Figure 1). We have shown that vascular patterning is extremely sensitive to the shape of endothelial cells. Thus the in-vitro technique can assay changes in cell shape resulting from small changes in cytoskeletal structure (in particular, changes in microtubule patterning). We are currently implementing this technique in the Glazier laboratory after Dr. Roeland Merks (a postdoc trained in it in the laboratory of Dr. Goligorsky. 8) We have continued to develop and refine our developmental-biology simulations (releasing the new open-source version of our code, CompuCell 3D). These simulations allow us to study the effects of phenomenological changes in cell behavior on macroscopic patterning. In 2004, our simulations primarily focused on chondrogenesis in the embryonic chick limb and in micromass culture and on HUVEC angiogenesis assays (See Figure 2). The purpose of these simulations is to help us interpret the causes of observed changes of tissue morphology in terms of changes in individual cell behavior, and to be able to predict the effects of changes at the cell level on macroscopic tissue behaviors (especially in angiogenesis and wound healing).

	9) In cooperation with co-PIs Dr. Lansford and Prof. Gabor Forgacs we have also conducted experiments measuring the adhesive properties of cell cultures exposed to simulated microgravity conditions. Glazier's graduate student, Wendy Zhang has made extended visits to both the Lansford and Forgacs Laboratories for training and collaborative research. We presented the following findings at the February NASA Microgravity Conference: We performed systematic comparative studies of developmental processes in 1g and simulated microgravity, using NASA's High Aspect Ratio Vessel (HARV). We focused on morphogenetic processes that both in vivo and in vitro result from the minimization of tissue surface and interfacial energies and proceed in analogy with liquid phenomena. We compared the rounding of irregular tissue explants into spheroids and measured the interfacial tensions of real tissues (embryonic heart cushions) and model tissues (constructed from Chinese Hamster Ovary (CHO) cells with controlled adhesive properties). We found a drastic reduction in tissue-medium interfacial tension (and thus tissue cohesivity) in simulated microgravity (80% reduction in CHO cell aggregates). The same reduction could be achieved by treating the tissues with the F-actin depolymerizing agent Latrunculin at 1 ?M concentration, suggesting that simulated microgravity severely impairs the cytoskeleton. We next considered cell sorting and the fusion of heart cushions during embryogenesis. We found that both sorting and cushion tissue fusion mimic the analogous process in liquid, the phase separation of immiscible liquids and the fusion of liquid drops, respectively. However, these morphogenetic transformations occur at a markedly different rate in simulated microgravity than in 1g. These findings imply that crucial early embryonic phenomena might be affected by altered gravity. More importantly, they suggest a way to maintain normal development even under reduced gravity. Ms. Zhang is currently working to adapt the Tracker viscometer in the Glazier laboratory to allow the Glazier laboratory to run equivalent adhesion assays for the early embryonic, mesenchymal and endothelial cells which the laboratory focuses on. Now that we have validated these techniques and have them running in the Glazier laboratory, in the 2005 calendar year we intend to apply them to the specific case of interest, the behavior of cells held under simulated microgravity conditions. We intend to use our single and tissue level assays to establish functional changes in these cells' motility, signaling response, adhesion and structure to evaluate possible health consequences of prolonged microgravity conditions. We have also received separate funding for a two-photon multispectral microscope for embryogenesis studies and have hired an expert microscopist (Dr. John Scott Gens) to conduct the experiments related to this grant.
Bibliography Type:	Description: (Last Updated: 12/29/2020)
Abstracts for Journals and Proceedings	Zeng W, Glazier JA, Weijer CJ. "Cell movement patterns during primitive streak initiation and extension in the chick resemble patterns in viscous flow." Society for Developmental Biology 63rd Annual Meeting, July 2004. Developmental Biology 2004 Jul;271: 574. , Jul-2004
Articles in Peer-reviewed Journals	Izaguirre JA, Chaturvedi R, Huang C, Cickovski T, Coffland J, Thomas G, Forgacs G, Alber M, Hentschel G, Newman SA, Glazier JA. "CompuCell, a multi-model framework for simulation of morphogenesis." Bioinformatics. 2004 May 1;20(7):1129-37. PMID: 14764549 , May-2004
Articles in Peer-reviewed Journals	Kiskowski MA, Alber MS, Thomas GL, Glazier JA, Bronstein NB, Pu J, Newman SA. "Interplay between activator–inhibitor coupling and cell-matrix adhesion in a cellular automaton model for chondrogenic patterning." Dev Biol. 2004 Jul 15;271(2):372-87. PMID: 15223341 , Jul-2004
Articles in Peer-reviewed Journals	Hentschel HGE, Glimm T, Glazier JA, Newman SA. "Dynamical mechanisms for skeletal pattern formation in the vertebrate limb." Proc Biol Sci. 2004 Aug 22;271(1549):1713-22. PMID: 15306292 , Aug-2004
Articles in Peer-reviewed Journals	Zeng W, Thomas GL, Glazier JA. "Non-Turing stripes and spots: a novel mechanism for biological cell clustering." Physica A: Statistical Mechanics and its Applications, 2004 Oct 1;341:482-94. http://dx.doi.org/10.1016/j.physa.2004.03.089 , Oct-2004
Articles in Peer-reviewed Journals	Merks RMH, Newman SA, Glazier JA. "Cell-oriented modeling of in vitro capillary development." Lecture Notes in Computer Science 2004;3305:425-34. http://dx.doi.org/10.1007/b102055 , Dec-2004
Articles in Peer-reviewed Journals	Chaturvedi R, Huang C, Izaguirre JA, Newman SA, Glazier JA, Alber MS. "A hybrid discrete-continuum model for 3-D skeletogenesis of the vertebrate limb." Lecture Notes in Computer Science 2004;3305:543-52. http://dx.doi.org/10.1007/b102055 , Dec-2004